PRESENTATION

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INHERITANCE OF
BASE CLASSES AND
DERIVED CLASES

INTRODUCTION

Inheritance is a fundamental concept of Object-Oriented Programming (OOP)

It allows one class to reuse properties and behavior of another class

Helps in code reusability and organization

WHAT IS INHERITANCE

Inheritance means acquiring properties of another class
The class whose properties are inherited → Base Class (Parent Class)
The class that inherits → Derived Class (Child Class)

BASE CLASS

A class that provides common properties and functions

Also known as Parent Class / Superclass

```
class Animal {
public:
    void eat() {
        cout << "Eating..." << endl;
    }
};
```

DARIVED CLASS

A class that inherits from the base class

Can use base class properties + add new features

```
class Dog : public Animal {
public:
    void bark() {
        cout << "Barking..." << endl;
    }
};
```

TYPES OF INHERITANCE

Single Inheritance

Multiple Inheritance

Multilevel Inheritance

Hierarchical Inheritance

Hybrid Inheritance

EXAMPLE

```
#include <iostream>
using namespace std;

class Animal {
public:
    void eat() {
        cout << "Animal is eating" << endl;
    }
};

class Dog : public Animal {
public:
    void bark() {
        cout << "Dog is barking" << endl;
    }
};

int main() {
    Dog d;
    d.eat();
    d.bark();
    return 0;
}
```

ADVANTAG AND DISADVAN TAGE

Code reusability
Reduces redundancy
Improves maintainability
Supports method overriding

Tight coupling between classes
Can increase complexity
Improper use may lead to errors

CONCLUSION

Inheritance is a powerful feature in OOP

Helps build structured and reusable code

Base class provides common functionality

Derived class extends and customizes behavior

SUBJECT 02

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